

SIMATS ENGINEERING

ASSESSMENT TOOL 2

**COURSE NAME: CRYPTOGRAPHY AND NETWORK
SECURITY COURSE CODE: CSA5113**

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I DINDU PUJITHA (192373037) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:D PUJITHA

Annexure A
Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

- A. Explain why this issue occurs.
- B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

Answer:

A. Explain why this issue occurs.

ECB (Electronic Codebook) mode encrypts each plaintext block independently using the same key. If two plaintext blocks are identical, they produce the same ciphertext blocks. As a result, repeated patterns in the original financial data remain visible in the ciphertext, making it easier for attackers to identify data patterns and reducing security.

B. Suggest a more secure block cipher mode and justify your choice.

Recommended Mode: CBC (Cipher Block Chaining)

Justification:

- CBC combines each plaintext block with the previous ciphertext block before encryption.
- It uses an Initialization Vector (IV), so identical plaintext blocks produce different ciphertext.
- It hides data patterns effectively and provides better confidentiality than ECB.
- Therefore, CBC is more suitable for securely transmitting repeated financial data.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

Answer:

A. Identify the most suitable algorithm for this scenario.

Recommended Algorithm: Blowfish

B. Justify your choice based on security strength and efficiency.

- Blowfish supports key sizes up to **448 bits**, providing strong security.
- It encrypts data faster than 3-DES while offering better security than DES.
- DES is no longer secure because its **56-bit key** can be broken using brute-force attacks.
- 3-DES is more secure than DES but is much slower.
- Blowfish offers a good balance between security and performance, making it suitable for banking applications.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

A. Explain how the key exchange works in this scenario.

B. Identify a possible security threat and suggest a solution to mitigate it.

Answer:

A. Explain how the key exchange works in this scenario.

In the Diffie-Hellman key exchange:

1. Both users agree on a public prime number (**p**) and generator (**g**).
2. Each user selects a private key.
3. Each user computes a public key and exchanges it over the public network.
4. Using their private key and the received public key, both users independently calculate the same shared secret key.
5. The shared secret is then used to encrypt future communications.

B. Identify a possible security threat and suggest a solution to mitigate it.

Security Threat: Man-in-the-Middle (MITM) Attack

An attacker can intercept the exchanged public keys and replace them with their own public keys. This allows the attacker to establish separate secret keys with each user and secretly read or modify the communication.

Solution:

- Use **Digital Certificates (PKI)**.
- Use **Digital Signatures** to authenticate public keys.
- Implement **Authenticated Diffie-Hellman** or **TLS/SSL** protocols.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

- A. Describe how RSA can be used for encryption and decryption in this scenario.
- B. Explain how key distribution and management play a role in maintaining security.

Answer:

A. Describe how RSA can be used for encryption and decryption in this scenario.

- Each user has a **public key** and a **private key**.
- The sender encrypts the email using the receiver's public key.
- The encrypted email is sent over the network.
- Only the receiver can decrypt the email using their private key.
- This ensures that only the intended recipient can read the message.

B. Explain how key distribution and management play a role in maintaining security.

- Public keys must be distributed securely so users can trust their authenticity.
- Private keys must be kept confidential and protected from unauthorized access.
- Digital certificates issued by trusted Certificate Authorities (CAs) help verify public keys.
- Proper key management, including secure storage and periodic key renewal, helps.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices.

B. Recommend a suitable cryptographic approach and justify your answer.

Answer:

A. Analyze the constraints of IoT devices.

IoT devices typically have:

- Limited processing power.
- Limited memory and storage.
- Low battery capacity.
- Limited computational resources.
- Need for fast and energy-efficient encryption.

These constraints make heavyweight cryptographic algorithms less suitable.

B. Recommend a suitable cryptographic approach and justify your answer.

Recommended Cryptographic Approach: Elliptic Curve Cryptography (ECC)

Justification:

- ECC provides the same level of security as RSA with much smaller key sizes.
- It requires less processing power and memory.
- It consumes less battery power, making it ideal for low-power IoT devices.
- ECC enables faster encryption and decryption while maintaining strong security.
- Therefore, ECC is the preferred choice for secure communication in smart home IoT systems.